

CS4084

Quantum Computing

Week 08 – Lecture 15

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Outline

- 1 Quick Review: What We have Covered Last Lecture
- 2 Projective measurements
- 3 Irrelevance of global phases
- 4 No-cloning theorem
- 5 Summary: Key Takeaways

Quick Review: What We Covered Last Lecture I

- **Quantum Circuits:** Graphical representation of quantum evolution.
 - **Wires** represent qubits; **Gates** represent unitary operations.
 - **Local vs Global:** Local gates act on single wires ($H \otimes I$); Global gates like CNOT operate across wires to enable **entanglement**.
- **Inner Products & Geometry:**
 - The inner product $\langle \phi | \psi \rangle$ defines the "overlap" between states.
 - **Orthogonality:** Two states are orthogonal if $\langle \phi | \psi \rangle = 0$.
 - **Orthonormality:** A basis $\{|v_i\rangle\}$ is orthonormal if $\langle v_i | v_j \rangle = \delta_{ij}$ (mutually orthogonal and unit length).
- **Projections:** Represented by $P = |\psi\rangle\langle\psi|$, used to project a state onto a specific subspace or basis vector.
- **Compound Systems & Tensor Products ($\otimes$):**
 - Joint states follow **lexicographical ordering**: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
 - **Multi-Qubit Space:** n qubits exist in $(\mathbb{C}^2)^{\otimes n}$.

Quick Review: What We Covered Last Lecture II

- **Matrix Tensor Product:** Joint operations use the **Kronecker product** ($M \otimes N$).
- **Bilinearity:** Tensor products are linear in every "slot," allowing scalars to move: $\alpha(|A\rangle \otimes |B\rangle) = (\alpha|A\rangle) \otimes |B\rangle$.
- **Probabilistic Operations:** Transformations on compound systems use **stochastic matrices** acting on the Cartesian product of state spaces.

Projective measurements

We can also choose different names for the measurement outcomes. Any collection of projections $\{\Pi_\alpha : \alpha \in \Gamma\}$ that satisfies the condition

$$\sum_{\alpha \in \Gamma} \Pi_\alpha = \mathbb{1}$$

describes a projective measurement having outcomes in the set Γ . The rules are the same as before:

1. The outcome $\alpha \in \Gamma$ of the measurement is chosen randomly:

$$\Pr(\text{outcome is } \alpha) = \|\Pi_\alpha |\psi\rangle\|^2$$

2. The state of the system becomes

$$\frac{\Pi_\alpha |\psi\rangle}{\|\Pi_\alpha |\psi\rangle\|}$$

Example

Standard basis measurements are projective measurements:

- The outcomes are the classical states of the system being measured.
- The measurement is described by the set $\{|a\rangle\langle a| : a \in \Sigma\}$.

Suppose that we measure the state

$$|\psi\rangle = \sum_{a \in \Sigma} \alpha_a |a\rangle$$

Each outcome a appears with probability $\| |a\rangle\langle a| \psi \|^2 = |\alpha_a|^2$.

Conditioned on the outcome a , the state becomes

$$\frac{|a\rangle\langle a| \psi}{\| |a\rangle\langle a| \psi \|} = \frac{\alpha_a}{|\alpha_a|} |a\rangle$$

Continue..

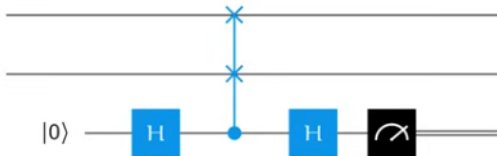
Example

Define two projections as follows:

$$\begin{aligned}\Pi_0 &= |\phi\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-| + |\psi^+\rangle\langle\psi^+| \\ \Pi_1 &= |\psi^-\rangle\langle\psi^-|\end{aligned}$$

The projective measurement $\{\Pi_0, \Pi_1\}$ is an interesting one...

Every projective measurements can be **implemented** using unitary operations and standard basis measurements.



Irrelevance of global phases

Definition

Suppose that $|\psi\rangle$ and $|\phi\rangle$ are quantum state vectors satisfying

$$|\phi\rangle = \alpha|\psi\rangle$$

The states $|\psi\rangle$ and $|\phi\rangle$ are then said to *differ by a global phase*.

(This requires $|\alpha| = 1$. Equivalently, $\alpha = e^{i\theta}$ for some real number θ .)

Suppose we apply a unitary operation to two states that differ by a global phase

$$U|\phi\rangle = \alpha U|\psi\rangle = \alpha(U|\psi\rangle)$$

They still differ by a global phase...

Consequently, two quantum state vectors $|\psi\rangle$ and $|\phi\rangle$ that differ by a global phase are *completely indistinguishable* and are considered to be *equivalent*.

Example..

Example

The quantum states

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \quad \text{and} \quad -|-\rangle = -\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

differ by a global phase.

Example..

Example

The quantum states

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad \text{and} \quad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

do **not** differ by a global phase. (This is a **relative phase** difference.)

This is consistent with the observation that these states can be discriminated perfectly:

$$\begin{array}{ll} |\langle 0 | H | + \rangle|^2 = 1 & |\langle 0 | H | - \rangle|^2 = 0 \\ |\langle 1 | H | + \rangle|^2 = 0 & |\langle 1 | H | - \rangle|^2 = 1 \end{array}$$

No-cloning theorem

Theorem (No-cloning theorem)

Let X and Y both have the classical state set $\{0, \dots, d-1\}$, where $d \geq 2$. There does not exist a unitary operation U on the pair (X, Y) such that

$$\forall |\psi\rangle : U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

The operation U must clone the standard basis states $|0\rangle$ and $|1\rangle$:

$$U(|0\rangle \otimes |0\rangle) = |0\rangle \otimes |0\rangle$$

$$U(|1\rangle \otimes |0\rangle) = |1\rangle \otimes |1\rangle$$

Therefore, by linearity,

$$U\left(\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle\right) = \frac{1}{\sqrt{2}}|0\rangle \otimes |0\rangle + \frac{1}{\sqrt{2}}|1\rangle \otimes |1\rangle$$

No-cloning theorem

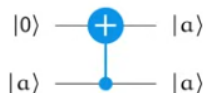
But this is not the correct behavior — we must have

$$\begin{aligned} U\left(\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle\right) \\ = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \end{aligned}$$

No-cloning theorem

Remarks:

- Approximate forms of the cloning theorem are known.
- Copying a standard basis state is possible — the no-cloning theorem does not contradict this.



- Cloning a probabilistic state (classically) is also impossible.

Summary: Key Takeaways I

- **Mathematical Foundations:**

- **Matrix Tensor Products:** Joint operations use the Kronecker product $(M \otimes N)$.
- **Bilinearity:** $\otimes$ is linear in every "slot," allowing scalars to move freely: $\alpha(|A\rangle \otimes |B\rangle) = (\alpha|A\rangle) \otimes |B\rangle$.

- **Multi-Qubit Geometry:**

- Multiple qubits live in the joint space $(\mathbb{C}^2)^{\otimes n}$.
- Independent states combine via $|\phi\rangle \otimes |\psi\rangle$ following **lexicographical ordering** (e.g., $|00\rangle, |01\rangle, |10\rangle, |11\rangle$).

- **Measurement & Projections:**

- **Projections:** Operators $P = |\psi\rangle\langle\psi|$ map states onto specific subspaces.
- **Projective Measurements:** The fundamental way we extract information, causing the state to "collapse" onto an eigenstate of the observable.

Summary: Key Takeaways II

- **Physical Realities:**

- **Irrelevance of Global Phases:** The state $e^{i\theta}|\psi\rangle$ is physically indistinguishable from $|\psi\rangle$; only *relative* phases matter.
- **No-Cloning Theorem:** A cornerstone of quantum security—it is impossible to create an identical copy of an arbitrary unknown quantum state.

- **Unitary Multi-Systems:**

- **Local:** $H \otimes I$ acts only on the first qubit.
- **Global:** Inseparable operations (like CNOT) are the engines of **entanglement**.

Next: Entanglement, Global Unitary Operations, and Multi-System Postulates.